

Tasks

1.

Use Cisco Packet Tracer to simulate a WAN setup connecting two branch offices via a VPN tunnel and configure NAT on the edge routers so both branches can access the internet.

Ans:

🔗 **Step 1:** Open Cisco Packet Tracer and set up two edge routers (Router-Branch1 and Router-Branch2) connected across a simulated ISP internet router.

🔗 **Step 2:** Configure the local LAN interfaces on both routers (Branch 1 LAN: 192.168.1.0/24, Branch 2 LAN: 192.168.2.0/24).

🔗 **Step 3:** Assign public IP addresses to the WAN interfaces facing the ISP router (Branch 1 WAN: 203.0.113.2, Branch 2 WAN: 198.51.100.2).

🔗 **Step 4:** Define the IPsec VPN parameters by setting up matching ISAKMP policies, a shared secret key, and crypto maps on both edge routers to secure the cross-branch traffic.

🔗 **Step 5:** Define the NAT boundaries on both routers by applying the ip nat inside command on local interfaces and ip nat outside on the internet-facing WAN interfaces.

2.

Configure Port Address Translation (PAT) on a router in your simulated WAN so that three devices from the internal network can browse the internet using a single public IP address.
Hint: Use the 'overload' keyword in your NAT configuration.

Ans:

🔗 **Step 1:** Connect three host PCs to the internal switch at Branch 1, assigning them private IP addresses 192.168.1.10, 192.168.1.11, and 192.168.1.12.

🔗 **Step 2:** Click on Router-Branch1, enter global configuration mode, and define an access list to grab the local subnet traffic:

```
Router-Branch1(config)# access-list 10 permit 192.168.1.0 0.0.0.255
```

Step 3: Link the access list to the public-facing WAN interface using the mandatory overload keyword to enable multi-device port sharing:

```
Router-Branch1(config)# ip nat inside source list 10 interface GigabitEthernet0/1 overload
```

🔗 **Step 4:** Open the web browser tool on all three internal PCs and navigate to a simulated public website IP address to trigger concurrent sessions.

🔗 **Step 5:** Run the verification command show ip nat translations on the router to watch three unique internal IPs share the single public WAN IP via distinct source port allocations.

3.

Intentionally misconfigure the VPN tunnel's pre-shared key between two routers in your WAN simulation, then use the 'show crypto isakmp sa' and 'show run' commands to identify and fix the connectivity issue.

Ans:

Step 1: Intentionally break the tunnel parameters by changing the key on Router-Branch2 to an incorrect value using the command:

```
Router-Branch2(config)# crypto isakmp key WRONGKEY address 203.0.113.2
```

🔍 **Step 2:** Try sending a ping packet from a host in Branch 1 to a host in Branch 2; notice that the connection fails and remains stuck.

🔍 **Step 3:** Open the CLI on Router-Branch1 and run the verification command show crypto isakmp sa to check the phase-1 tunnel status.

🔍 **Step 4:** Identify the problem by analyzing the output state, which will display MM_NO_STATE or remain blank, confirming that the security keys do not match.

🔍 **Step 5:** Review the configurations line by line using the show running-config command on both routers to isolate the key spelling mismatch.

🔍 **Step 6:** Fix the error on Router-Branch2 by re-entering the correct matching pre-shared key:

```
Router-Branch2(config)# crypto isakmp key CorrectKey address 203.0.113.2
```

4.

Troubleshoot a scenario where devices in one branch office can ping the internet but cannot access devices in the other branch via the VPN tunnel; document the commands you use and the configuration change that resolves the problem.

🔍 **Step 1:** Identify the issue where PCs can ping the internet but cannot access the remote branch; this indicates that the branch-to-branch traffic is being accidentally intercepted and altered by the NAT engine.

🔍 **Step 2:** Run the command show access-lists on the edge router to view the current NAT rule parameters.

🔍 **Step 3:** Notice that the NAT rule grabs all local traffic and forces it to translate, without adding an exception rule for the private remote branch network.

🔍 **Step 4:** Fix the issue by modifying the NAT Access Control List to explicitly block branch-to-branch data from being translated before matching the general internet traffic:

```
Router-Branch1(config)# access-list 10 deny ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
```

```
Router-Branch1(config)# access-list 10 permit ip 192.168.1.0 0.0.0.255 any
```

Step 5: Execute the clear ip nat translation * command to refresh the path, and confirm that cross-branch communication is restored through the secure tunnel.

5.

Take the WAN, VPN, and NAT scenario you built and swap your configuration file with a classmate; review their setup, find one misconfiguration, and write a short note on how you identified and fixed it.

Constraint: Focus on using 'show' and 'debug' commands for troubleshooting rather than guessing the issue.

Ans:

🔍 **How I Identified the Issue:** I imported a classmate's configuration file into Packet Tracer. Instead of guessing, I opened the CLI and typed `show crypto map` and noticed that the tunnel crypto map was properly configured but not bound to any physical port. To double-check, I ran the active monitoring command `debug crypto isakmp`, triggered a ping, and saw that no VPN negotiation packets were actively leaving the WAN interface.

🔍 **How I Fixed It:** I typed `show running-config` to verify the exact name of the configured crypto map (which was VPN-MAP). I then accessed the global configuration mode, stepped into the public WAN interface port, and manually activated the map using the following commands to instantly fix the connection issue:

```
Router(config)# interface GigabitEthernet0/1
```

```
Router(config-if)# crypto map VPN-MAP
```